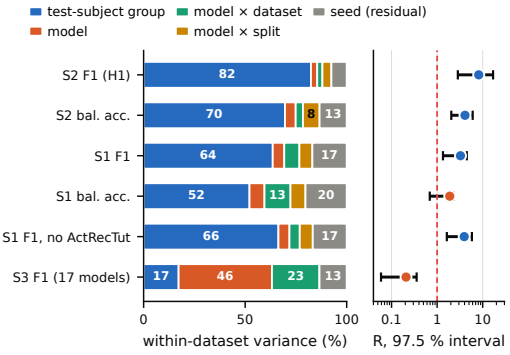
How Much of Wearable Activity-Recognition Accuracy Is the Network? A Pre-Registered Variance Decomposition Under Person-Disjoint Evaluation

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Abstract—Wearable activity-recognition systems are compared by macro-F1 on people unseen in training, but how much of that figure reflects the network rather than the people tested is unknown. We decompose the variance of stored runs of one person-disjoint protocol (WHAR Arena v1) with a linear mixed model and a pre-registered test. On 7 held-out datasets (828 runs of four networks, three seeds), the test-subject group accounts for 82 % of the within-dataset variance, 8.19 times (97.5 % interval 2.85–16.90) all network-related variance; the pre-registered criterion was met. The ordering holds with a ceiling-free metric and on development datasets, where that metric leaves it unresolved, and it reverses across 17 heterogeneous published architectures. For comparable networks, reported accuracy depends more on who is tested than on the network: comparisons must be paired by split and report the spread across subjects.



Index Terms—Wearable sensors, inertial sensors, human activity recognition, person-disjoint evaluation, variance components.

I. INTRODUCTION

Wearable inertial sensing systems for human activity recognition (HAR) are designed and chosen on one number: macro-F1 on subjects held out from training [1]. Evaluating on unseen people removes the optimism of random cross-validation [2], and how people are partitioned between training and test data changes the reported accuracy [3]. People move differently [4] and devices sense differently [5], so a test set of a few subjects carries its own variation. WHAR Arena [6] fixed one person-disjoint protocol across 30 datasets and found the state of the art distributed over architectures rather than held by one; it reports one training run per split, so it cannot separate the effect of who is tested from the training seed. Machine-learning benchmarks are known to depend on such sources of variation [7], but their sizes in wearable HAR are unknown.

This letter measures them. Under one fixed protocol it decomposes the variance of macro-F1 into the group of test subjects, the network, their interactions and the training seed; it tests the ordering of these terms with a pre-registered analysis on held-out datasets; it checks the result with a ceiling-free metric, development data and the benchmark's independent architectures; and it reports, as exploratory, which design choices and which sensing properties of a dataset go with large accuracy differences, what the networks' temporal memories learned, and how far the test subjects lie from the training subjects in the networks' embedding space.

II. METHODS

A. Datasets and sensors

The 18 datasets of the stored runs (Table 1, Fig. 1) were recorded with smartphones, body-worn inertial nodes, a smartwatch, smart glasses and a helmet. Every dataset has accelerometers; most add gyroscopes, several magnetometers, and single datasets a barometer, an ambient-light sensor, stretch sensors, ECG or EEG. They span 3 to 133 channels, 2 to 36 subjects, 3 to 27 classes and 20 to 250 Hz, with 232,254 windows in all. Devices and modalities are quoted from the benchmark's dataset table [6]; channel and class counts, sampling rates and window counts are read from the stored runs, and subject counts from the harness's splits (for USC-HAD and MotionSense, whose caches we do not hold, from the dataset configurations).

B. Protocol and stored runs

All runs follow WHAR Arena v1 with its paper-era code [6], [8]: 3-s windows with 50 % overlap at the native sampling rate; $k = 10$ subject groups, each split testing on one group (1 to 4 subjects) that never appears in training or validation; validation from 20 % of the training subjects' windows; a z-score fitted on the training split; macro-F1 over all classes, with classes absent from a test split counted as 0. Apart from a label-permutation control, no network was trained for this letter: we analyse stored runs and their checkpoints. Three sets enter (Table 3): S1, ten networks that share one skeleton and differ in single components (with and without temporal memory, e.g. S4D [9], [10] and Mamba [11], or window statistics), on ten development datasets with three seeds; S2, four of these networks on eight datasets never used for this question, with three seeds; S3, the benchmark's 17 published models (deep networks such as DeepConvLSTM [12], [13], and Random Forest [14], SVM and k-NN on tsfresh features [15]) re-run with the same code on four datasets, one run per split. A pre-registered inclusion rule removed datasets whose stored runs disagree on split sizes, because their split indices do not name the same subjects across runs (one dataset each in S1 and S2). Fig. 2 summarises the design, the runs analysed and how subjects are assigned to test groups.

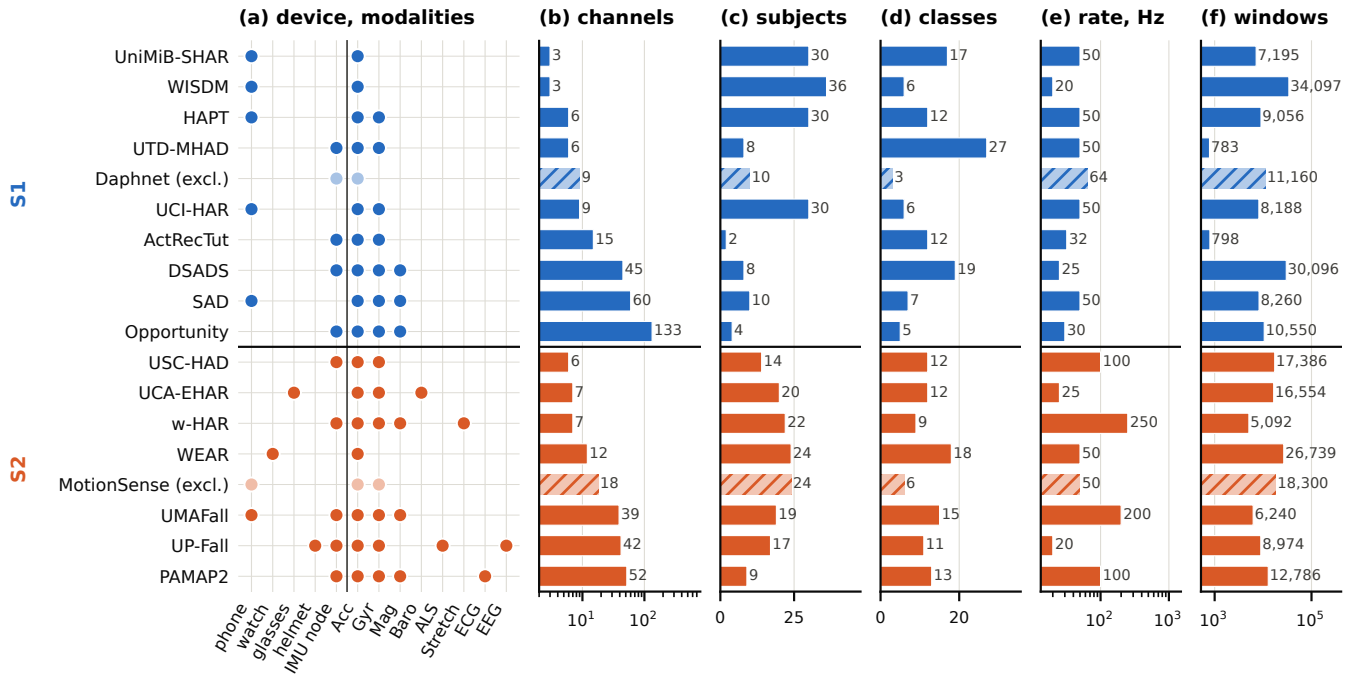


Fig. 1. Sensor configuration of the datasets of S1 (development) and S2 (held-out). (a) Devices and sensor modalities, quoted from the benchmark's dataset table [6]; (b)–(f) channels, subjects, classes, sampling rate and number of 3-s windows (logarithmic axes in (b), (e) and (f)), read from the stored runs and the harness's splits. Faded and hatched: excluded from the variance analyses by the pre-registered inclusion rule.

TABLE 1. Datasets of the stored runs

Dataset	Set	Device	Modalities	C	Subj.	Cl.	Hz	Windows	Splits
UniMiB-SHAR	S1	phone	Acc	3	30	17	50	7,195	10
WISDM	S1	phone	Acc	3	36	6	20	34,097	10
HAPT	S1	phone	Acc, Gyr	6	30	12	50	9,056	10
UTD-MHAD	S1	node	Acc, Gyr	6	8	27	50	783	8
Daphnet†	S1	node	Acc	9	10	3	64	11,160	10
UCI-HAR	S1	phone	Acc, Gyr	9	30	6	50	8,188	10
ActRecTut	S1	node	Acc, Gyr	15	2	12	32	798	2
DSADS	S1	node	Acc, Gyr, Mag	45	8	19	25	30,096	8
SAD	S1	phone	Acc, Gyr, Mag	60	10	7	50	8,260	10
Opportunity	S1	node	Acc, Gyr, Mag	133	4	5	30	10,550	4
USC-HAD	S2	node	Acc, Gyr	6	14	12	100	17,386	10
UCA-EHAR	S2	glasses	Acc, Gyr, Baro	7	20	12	25	16,554	10
w-HAR	S2	node	Acc, Gyr, Mag, Stretch	7	22	9	250	5,092	10
WEAR	S2	watch	Acc	12	24	18	50	26,739	10
MotionSense†	S2	phone	Acc, Gyr	18	24	6	50	18,300	10
UMAFall	S2	node, phone	Acc, Gyr, Mag	39	19	15	200	6,240	10
UP-Fall	S2	node, helmet	Acc, Gyr, ALS, EEG	42	17	11	20	8,974	10
PAMAP2	S2	node	Acc, Gyr, Mag, ECG	52	9	13	100	12,786	9

C: channels; Subj.: subjects; Cl.: classes; Hz: sampling rate; Windows: 3-s windows, each in one test group; Splits: test-subject groups. †Excluded from the variance analyses by the pre-registered inclusion rule.

C. Networks

The ten networks of S1 and S2 share one skeleton (Fig. 3, Table 2). A linear encoder maps the C channels of each sample to D features, N residual blocks process the sequence, and the features are averaged over the window, joined in four networks by a window-level side path, normalised (LayerNorm) and classified by a linear head. The vector the head receives is the network's embedding (Section II-F). In a residual block, a LayerNorm and a linear map give per-channel signals u that feed a memory; u and the memory's outputs are concatenated, mixed linearly and passed through a GELU into the residual sum. RecurrenceNet's memory keeps soft rows of a recurrence plot [16] over ten lags: whether a learned two-dimensional projection of the signal is back where it was τ samples earlier. HysteresisNet keeps four play operators per channel, whose output moves only when the input leaves a band of half-width r around it [17] (here with input-dependent widths and a clock channel). QuantileNet keeps four online quantile trackers per channel, stochastic approximation steps [18] that settle on a running quantile of level τ. The no-

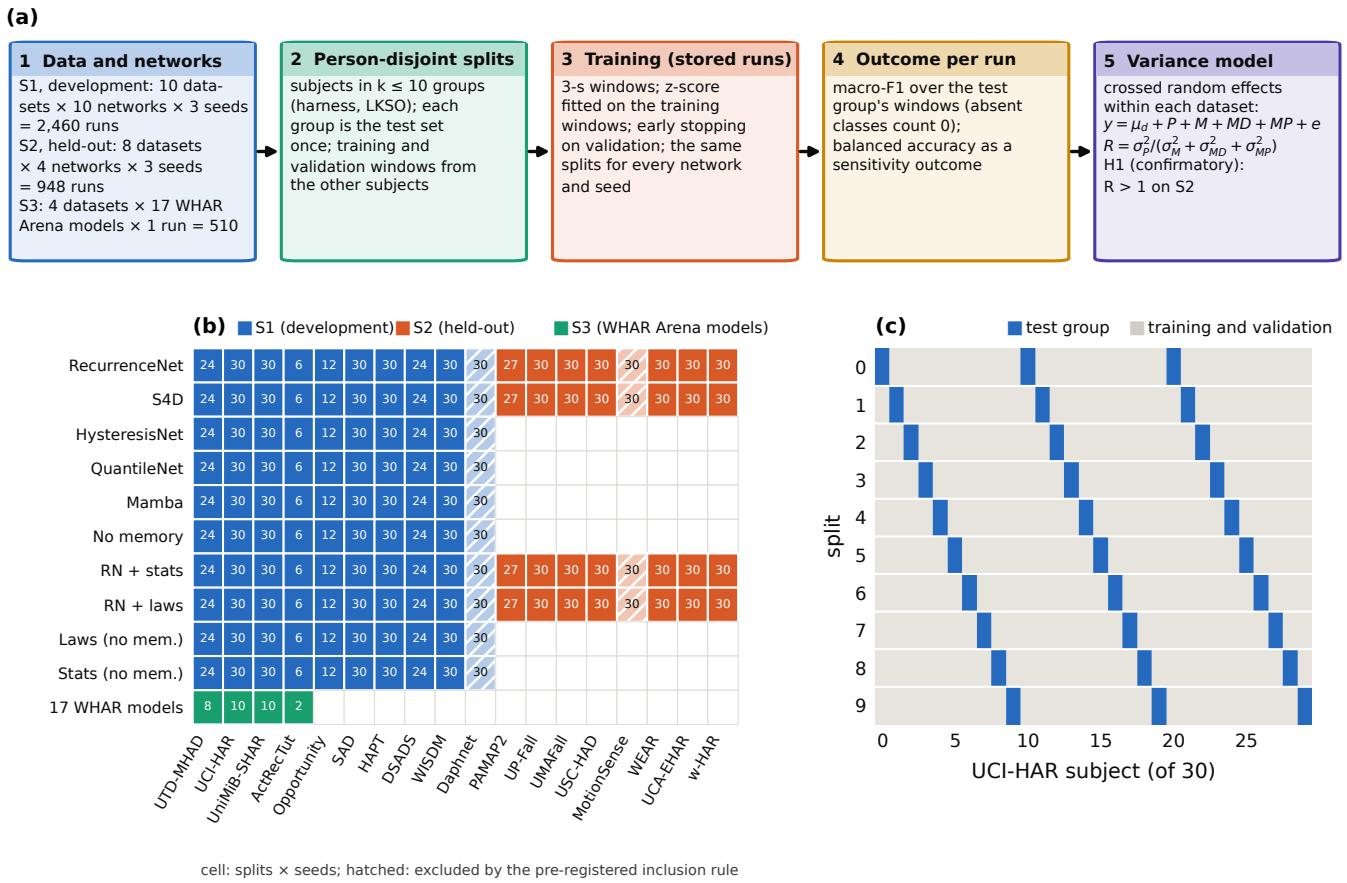


Fig. 2. Study design. (a) From the stored runs to the variance decomposition. (b) The runs: networks (rows) by datasets (columns), coloured by run set; each cell gives splits × seeds, and hatched cells are excluded by the inclusion rule. (c) The person-disjoint test groups of UCI-HAR: each split tests on three subjects that never enter its training or validation data.

TABLE 2. The ten networks

Network	Memory	Side path	Sets	$D \times N$	Emb.	Params (ref.)	Range over datasets
RecurrenceNet	recurrence	none	S1, S2	32×3	32	24,675	23,886–28,013
S4D	S4D (diagonal SSM)	none	S1, S2	64×2	64	27,675	26,118–34,373
HysteresisNet	play operators	none	S1	32×3	32	23,579	22,790–26,917
QuantileNet	quantile trackers	none	S1	32×3	32	20,763	19,974–24,101
Mamba	Mamba (S6)	none	S1	32×2	32	21,147	20,358–24,485
No memory	none (play width 0)	none	S1	32×3	32	19,995	19,206–23,333
RN + stats	recurrence	channel stats	S1, S2	32×3	64	26,451	24,582–46,357
RN + laws	recurrence	law invariants	S1, S2	32×3	64	27,553	25,316–49,437
Laws (no mem.)	none (plain block)	law invariants	S1	48×3	80	19,033	16,412–42,597
Stats (no mem.)	none (plain block)	channel stats	S1	48×3	80	17,931	15,678–39,517

Emb.: embedding dimension. Params (ref.): with the reference input of gate G1; range over the datasets of S1 and S2.

memory control sets every play width to zero, which makes each operator the identity, and the two networks without memory use plain blocks. S4D [10] and Mamba [11] use their own blocks, a diagonal state-space convolution and a selective scan. The side paths read per-channel window statistics (mean, standard deviation, minimum and maximum, the kind of features behind the Random Forest) or invariants of each three-axis sensor's temporal operator and of the couplings between sensors ("laws"). With the reference input of gate G1 (that of UTD-MHAD), the networks have 17,931 to 27,675 parameters, and on the datasets of S1 and S2 15,678 to 49,437.

D. Variance model and pre-registered test

For run (m, d, s, r) of network m on split s of dataset d with seed r,

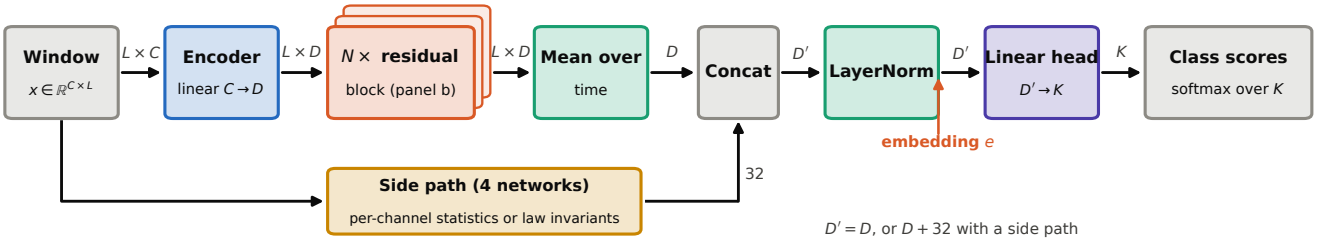
$$y_{mdsr} = \mu_d + P_{ds} + M_m + (MD)_{md} + (MP)_{mds} + e_{mdsr}, \quad (1)$$

with μ_d fixed and the other terms independent zero-mean random effects, fitted by REML [19] in statsmodels [20]. The estimand is $R = \sigma_P^2/(\sigma_M^2 + \sigma_{MD}^2 + \sigma_{MP}^2)$, the variance of the test-subject group over all network-related variance; π_P is σ_P^2's share of the within-dataset variance. Intervals are percentile 97.5 % intervals from 1000 cluster-bootstrap resamples that draw splits within datasets [21]. Before any variance analysis of S2, we registered H1 ($R > 1$ on S2), with success if the interval's lower bound exceeds 1 and failure if $R \leq 1$, sealed the analysis code by hash, and logged S2's single read. S2's per-dataset means were known from an earlier study; its variance components were not. For S3 (one run per split) the interaction and the seed cannot be separated, so the conservative $R' = \sigma_P^2/(\sigma_M^2 + \sigma_{MD}^2 + \sigma_e^2)$ is reported.

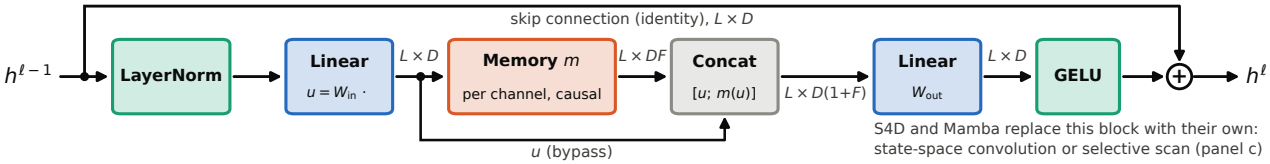
E. Exploratory analyses

Component contrasts on S1 pair two networks by split, average the seeds, and summarise over datasets with a t-interval and a Wilcoxon test [22], [23]. Over the benchmark's 30 datasets we regress the published single-run results on standardised dataset properties

(a) Shared skeleton of the ten networks



(b) Residual block $\ell = 1, \ldots, N$ (recurrence, play, quantile and plain blocks)



(c) Memories: update per channel and time step t (input u_t)

Memory	Update rule	State per channel	Networks
Recurrence plot	$r_t(\tau) = \sigma((\varepsilon - \lVert z_t - z_{t-\tau} \rVert)/T)$ and $r_t(\tau)\, r_{t-1}(\tau)$, z_t: 8 learned 2-D projections of u_t; 10 lags $\tau \le 48$	48-sample buffer	RecurrenceNet, RN + stats, RN + laws
Play operator	$y_t = \min(\max(y_{t-1}, u_t - r), u_t + r)$, 4 widths r (HysteresisNet: r depends on u_t, plus a clock channel)	4 values y	HysteresisNet; No memory ($r = 0$)
Quantile tracker	$q_t = q_{t-1} + \eta(\tau - \mathbf{1}[u_t < q_{t-1}])$, 4 levels τ	4 values q	QuantileNet
Diagonal SSM	$s_t = \bar{A}\, s_{t-1} + \bar{B}\, u_t$, $y_t = \mathrm{Re}(C\, s_t) + D u_t$; FFT convolution	16 complex modes	S4D
Selective scan	$s_t = \bar{A}_t\, s_{t-1} + \bar{B}_t\, u_t$, $y_t = C_t s_t$; Δ_t, B_t, C_t computed from u_t	16 states	Mamba
None	plain block: the memory and the concatenation are absent	none	Laws, Stats (no mem.)

Fig. 3. The networks of S1 and S2 as a dataflow diagram. (a) The shared skeleton, with the tensor shapes on its edges (C channels, L samples, width D, K classes); the embedding analysed in Section II-F is the head's input. (b) The residual block with its skip connection; the memory returns F features per channel. (c) Each memory's update per channel and time step, the state it keeps, and the networks that use it (sizes in Table 2).

(ordinary least squares); these associations are correlational, and their p-values are not corrected for multiplicity. Within each dataset of S1 and S2 we compare the spread of the split means (over networks and seeds) with that of the network means (over splits and seeds). From the stored checkpoints of eight S1 datasets we read what the memories learned and compare it with the same networks freshly initialised: S4D's mode time constants and decay rates, the play widths, the quantile levels and steps, and the recurrence radii. None of these analyses was pre-registered.

F. Embedding analysis

This analysis is exploratory and uses inference only. For the seed-0 checkpoint of every split of the 11 datasets whose caches on our cluster reproduce the stored runs (eight of S1 with the ten networks, three of S2 with four), the harness rebuilds the split and the network embeds every training and test window; the regenerated macro-F1 equals the stored one in 686 of 754 cases (largest difference 0.20 points). Three quantities describe each network and split. First, the class-conditional shift of the test group [24]: for every class present in both, the distance between the test and the training class means, in units of the training class's root-mean-square spread per dimension, averaged over classes. The same shift is computed in a sensor-feature space that no network shapes (per-channel mean, standard deviation, minimum and maximum of the normalised window). Second, the subject information in the training embeddings: the balanced accuracy of a linear probe [25] (logistic regression) that predicts the training subject, in five folds grouped by recording session so that overlapping windows never straddle folds, on at most 4,000 windows, against chance (one over the number of training subjects). Third, the same probe for the activity. Within each dataset we rank-correlate the test group's shift, averaged over networks, with the split's macro-F1 (Spearman), and we pool the within-dataset ranks over datasets with more than three splits. One case, fixed before the analysis (UCI-HAR, S4D, split 0), is mapped to two dimensions with t-SNE [26].

G. Verification

For verification, the benchmark's Random Forest, re-run with our code, reproduces its published macro-F1 on four datasets (e.g. 84.69 against 84.69 on UCI-HAR); test predictions were regenerated from 408 checkpoints, 373 of which reproduced the stored metrics exactly and all within two test windows; an independent REML implementation and negative controls (e.g. permuting network labels within cells) confirmed the analysis; and S4D retrained on permuted training labels (6 runs on UP-Fall and UCA-EHAR) fell to chance,

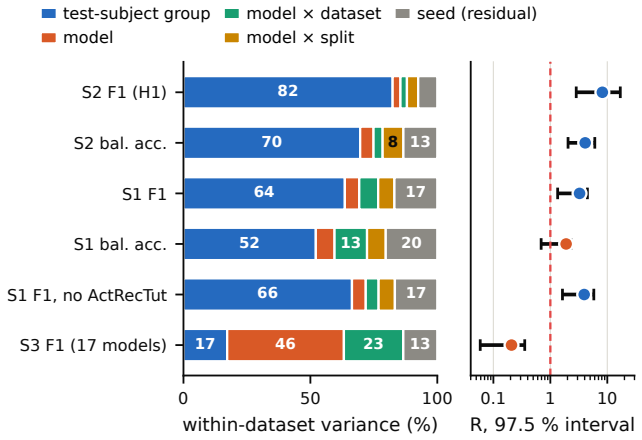


Fig. 4. Where the within-dataset variance of the stored runs lies. Left: shares by source (numbers in %; the seed term, grey, is the residual). Right: R with its 97.5 % bootstrap interval, blue where the interval lies above 1 and orange otherwise; the dashed line is the pre-registered threshold. R exceeds 1 in every analysis of the shared-skeleton networks (on S1 with balanced accuracy its interval includes 1), and the network dominates across the 17 heterogeneous published architectures (S3, conservative R').

with a test macro-F1 of 4.9 against 8.7 by chance and 54.5 with the true labels. Of the 828 confirmatory runs, the gate regenerated 360 on the caches the stored runs used, within that tolerance. Beyond the gate, every analysed run with a checkpoint was re-executed (2,982 of 2,988): 2,185 reproduce their records exactly, 2,668 within two test windows, and none differs by more than 3.25 F1 points. The residual differences have one source. Several parsers number activities and subjects, and order recordings, in the order the file system lists them, and many stored runs were built on cloud machines whose listing order cannot be recreated. For DSADS we recovered that order from the checkpoints themselves, since each network is near-perfect on its own training subjects, which identifies its output labels and its test subjects, and rebuilt the cache accordingly; for USC-HAD and three datasets whose caches we rebuilt, the test sets are identical and only the draw of validation windows, and with it the train-only z-score, differs. The 6 runs without a checkpoint were retrained to obtain one; 5 of them reproduce their record's macro-F1 exactly, and the stored records remain the ones analysed.

III. RESULTS

A. Confirmatory test

On S2 the test-subject group dominates (Fig. 4, Table 3): σ^2_P = 127.7 (F1 points^2) against σ^2_M = 4.8, σ^2_{MD} = 4.0 and σ^2_{MP} = 6.9, with the seed at σ^2_e = 11.7. Hence R = 8.19 [2.85, 16.90] and π_P = 82 % [64, 89]: the pre-registered decision is *success*, and the secondary criterion (π_P lower bound ≥ 40 %) is met.

B. Sensitivity analyses

Balanced accuracy, which averages recall over the classes present and so has no per-split ceiling, gives R = 4.09 [2.05, 6.03] on S2: ceilings enlarge the subject share but do not create it. On the development set S1 the ordering is the same with macro-F1 (R = 3.26 [1.34, 4.47]), also without its two-subject dataset ActRecTut (R = 3.93); with balanced accuracy the point estimate stays above 1 but the interval does not exclude it (R = 1.89 [0.69, 2.29]). Across the benchmark's 17 published architectures it reverses: R' = 0.21 [0.06, 0.36], with σ^2_M = 100.2 against σ^2_P = 37.5. These architectures range from k-NN to attention networks; their differences are far larger than those between the shared-skeleton networks.

TABLE 3. Variance components (F1 points^2) and R [97.5 % interval]

Analysis	n	σ^2_P	σ^2_M	σ^2_{MD}	σ^2_{MP}	σ^2_e	R [97.5 % int.]
S2 F1 (H1)	828	127.7	4.8	4.0	6.9	11.7	8.19 [2.85, 16.90]
S2 bal. acc.	828	86.3	6.5	4.4	10.1	16.6	4.09 [2.05, 6.03]
S1 F1	2160	66.2	5.9	7.8	6.7	17.6	3.26 [1.34, 4.47]
S1 bal. acc.	2160	39.2	5.5	9.6	5.5	15.3	1.89 [0.69, 2.29]
S1 F1, no ActRecTut	2100	67.4	5.5	5.1	6.5	17.0	3.93 [1.63, 5.80]
S3 F1 (17 models)	510	37.5	100.2	51.0	–	29.2	0.21 [0.06, 0.36]

C. Spread across test groups and networks

The decomposition's result is visible in the raw means (Fig. 5a). In 15 of 16 datasets of S1 and S2 the split means vary more than the network means, by a median factor of 2.6 in standard deviation; the exception is ActRecTut, whose two splits each test one of its two subjects. On UCI-HAR the split means have a standard deviation of 3.0 F1 points against 1.5 for the network means, and on UP-Fall 8.3 against 4.4. In the network × split cells (Fig. 5b, c) the columns dominate: a test group that is hard for one network is hard for the others.

D. Validation on the training subjects

Each run's validation windows come from its training subjects, so its validation score measures accuracy on people the network has seen (Fig. 6). On the 13 datasets whose runs all recorded it, the validation macro-F1 exceeds the test macro-F1 in 2,307 of 2,388 runs, by a median of 17.1 points per dataset (from 4.6 to 47.8); the four largest gaps are on held-out datasets. Validation is also a weak guide to which network generalises: the rank correlation between the networks' validation and test means has a median of +0.18 over datasets, and the network best on validation is the best on test in 3 of 13 datasets (a median loss of 2.7 test points when chosen by validation).

E. Exploratory findings

Within S1, the one design choice worth several points is temporal memory (Fig. 7a): S4D gains +5.79 [+2.30, +9.27] points over the same network without memory (9 of 9 datasets), while the type of memory moves macro-F1 by at most a few points (Mamba – S4D: −2.45 [−4.40, −0.50]). Over the 30 datasets, a larger channel count goes with a larger margin of the feature-based Random Forest over the best deep network (+7.1 points per standard deviation of $\log_2$ channels, $p < 0.001$, other properties held fixed; Fig. 7b), and fewer training subjects go with a larger gap between the best and the median deep network (−1.5 per standard deviation, p = 0.004). No property predicts how hard a dataset is (R^2 = 0.03).

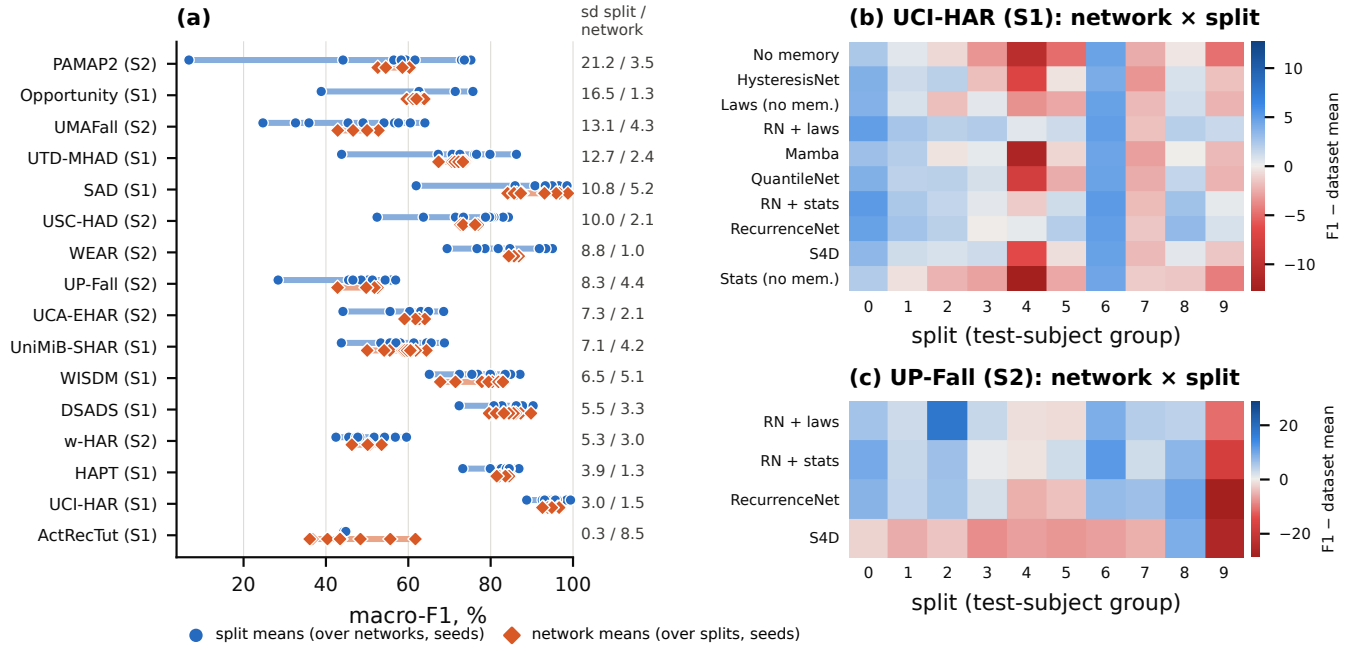


Fig. 5. Exploratory. (a) Within each dataset of S1 and S2, the means of the splits (circles; over networks and seeds) and of the networks (diamonds; over splits and seeds); right, the standard deviation of each. (b, c) Network × split cell means (over seeds) of UCI-HAR and UP-Fall, as deviations from the dataset mean.

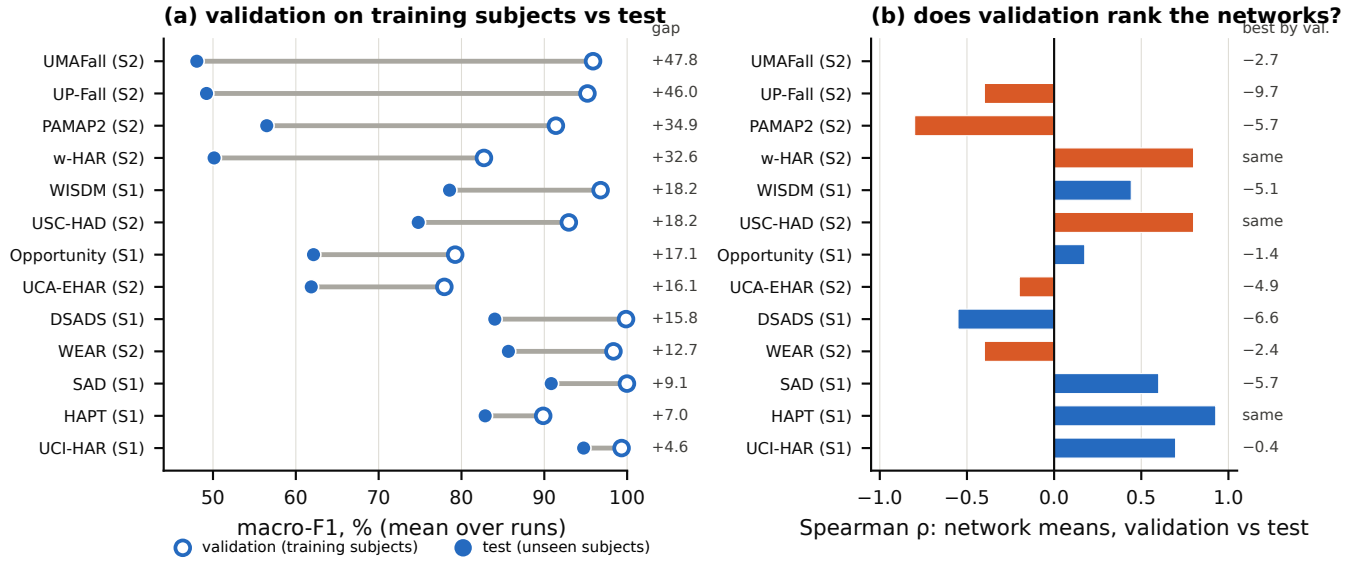


Fig. 6. Exploratory. (a) Per dataset, the mean validation macro-F1 (windows of the training subjects) and the mean test macro-F1 (unseen subjects) of the stored runs, and their gap. (b) Spearman correlation between the networks' validation and test means (S1 blue, S2 orange); right, whether the network best on validation is also best on test, or how many test points the choice loses. Datasets whose runs did not all record a validation score are omitted.

F. What the memories learned

The memories' own parameters barely move in training (Fig. 8; 766 checkpoints of eight S1 datasets, each against the same network freshly initialised). The quantile levels change by a median of 0.0047 (95th percentile 0.041), the recurrence radii by 0.014 (0.057), and S4D's decay rates by a median of 0.028 in $\log_{10}$. The play widths (median 0.36, initially 0.35) and S4D's mode time constants (median 1.8 window lengths, initially 1.6; 62 % longer than the window, initially 60 %) keep the distributions they start with. The time and amplitude scales at which each memory works are thus fixed by its design and initialisation; whatever memory adds (Fig. 7a), it adds at the scales it was given.

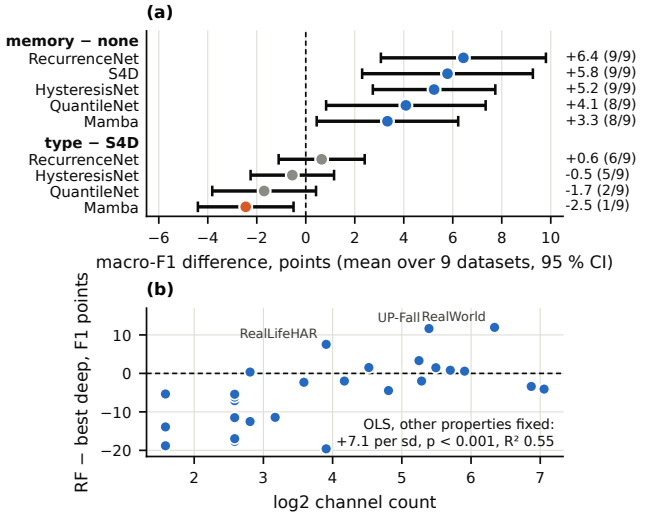


Fig. 7. Exploratory. (a) Temporal memory against none, and memory types against S4D, over the development datasets whose runs pair by subject (mean and 95 % CI, blue where the CI lies above 0, orange below 0, grey across 0; datasets won). (b) Over the benchmark's 30 datasets (published single runs), the Random Forest's margin over the best deep network rises with the number of sensor channels.

G. Embeddings

In the networks' embedding space, the hard test groups are the distant ones (Fig. 9). Within datasets, the splits whose test subjects lie further from the training subjects, class by class, are the splits on which the networks score lower: the pooled within-dataset rank correlation is $\rho = -0.46$ ($p < 0.001$), negative in 10 of 10 datasets (median -0.61) and, for single networks, in 69 of 82 network–dataset pairs (median -0.50). Measured per embedding dimension, so that networks of different width weigh alike, it is $\rho = -0.45$. The same shift measured on window statistics of the normalised signal, which no network shapes, tracks accuracy less closely ($\rho = -0.32$, $p = 0.002$; negative in 9 datasets). The training embeddings still carry the subject's identity: a linear probe identifies the training subject at a median 3.2 times chance, against 4.0 times from the window statistics, while the activity probe reaches a median balanced accuracy of 99 %. The networks thus reduce the subject information in their features without removing it. These associations are exploratory and correlational.

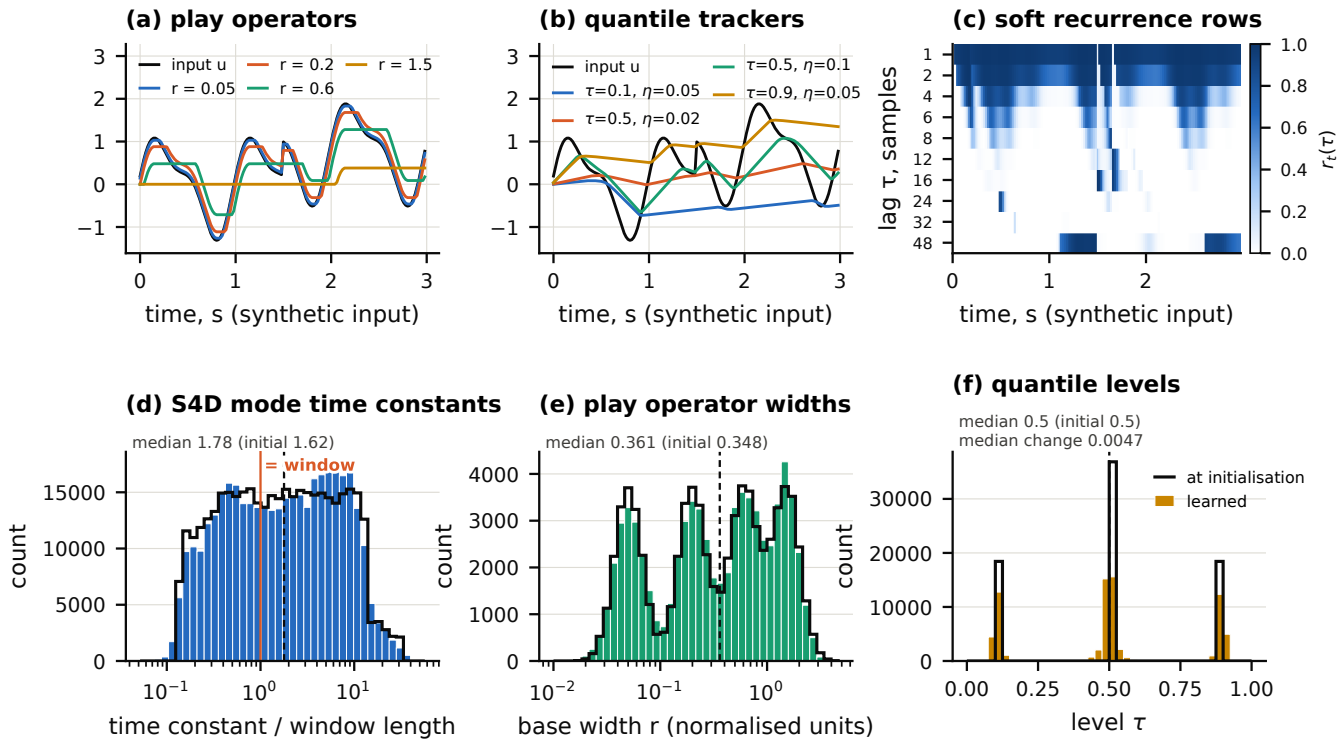


Fig. 8. The temporal memories. (a–c) The operators of HysteresisNet, QuantileNet and RecurrenceNet at their initial settings on a synthetic 3-s input (an illustration, not data). (d–f) Their parameters after training (bars) against the same networks freshly initialised (outline), over all stored checkpoints of eight S1 datasets: S4D's mode time constants in window lengths, the play widths, and the quantile levels.

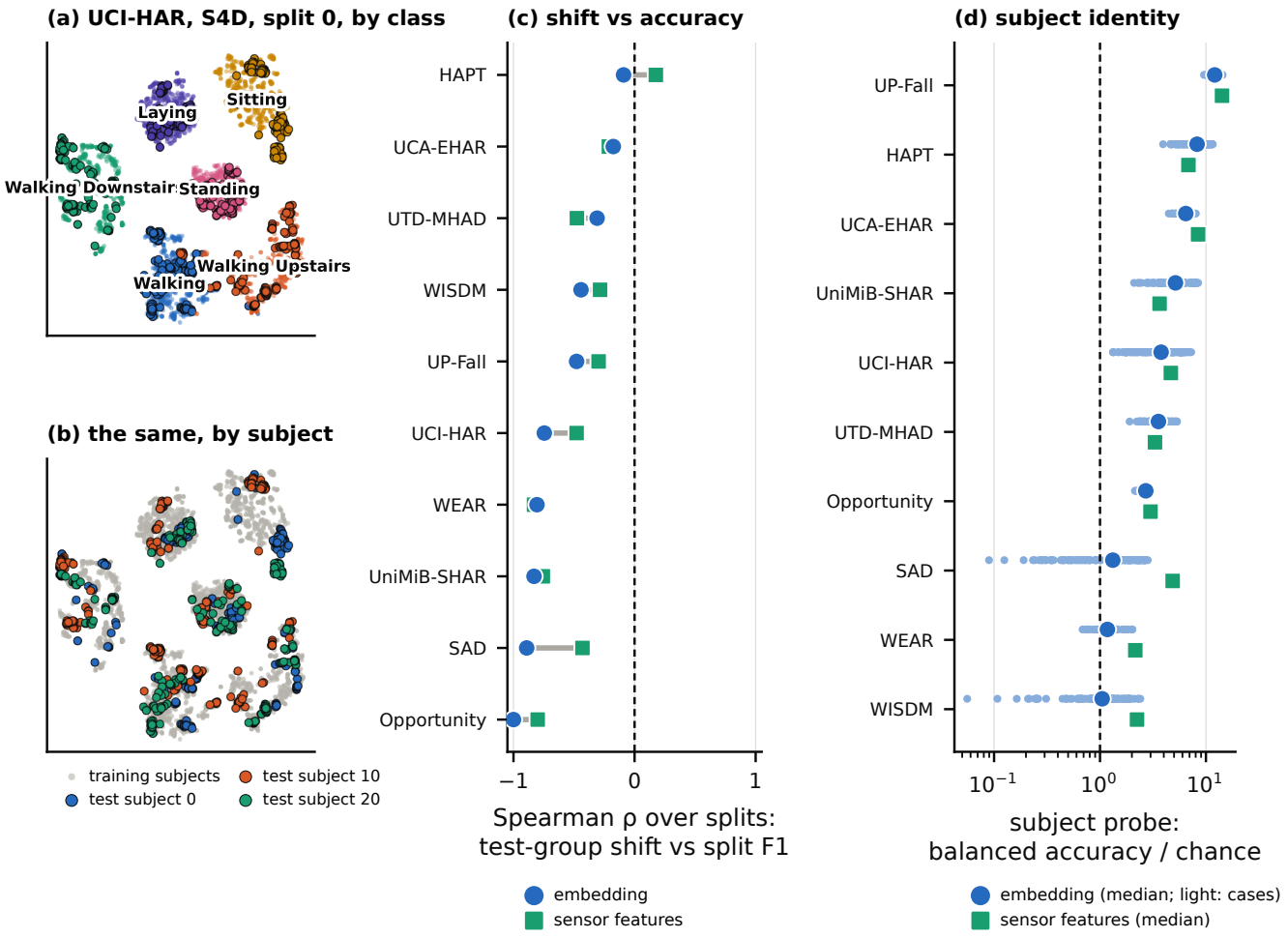


Fig. 9. Exploratory embedding analysis (seed-0 checkpoints, inference only). (a, b) t-SNE of the embeddings of one case fixed in advance (UCI-HAR, S4D, split 0): training windows (small, faded; a seeded subsample) and the test windows of the three unseen subjects (outlined), coloured by activity (a) and by subject (b). (c) Per dataset, Spearman correlation over splits between the test group's class-conditional shift (mean over networks) and the split's macro-F1, in the embedding and in a sensor-feature space. (d) Balanced accuracy of a linear probe that identifies the training subject, over chance, from the embeddings (all cases and median) and from the window statistics (median).

IV. DISCUSSION AND LIMITATIONS

Here "comparable" networks share one design and differ in single components, so that their dataset means differ by a few points. For such networks, who is tested decides more of a reported macro-F1 than which network is used. The variance between test-subject groups governs the absolute level, and with it every comparison made across studies, splits or subject groups. A comparison paired by split cancels it, but is then limited by the network × split interaction and the seed, which on S2 are as large as the network terms themselves (Table 3). For a wearable sensing study this gives three rules. Record enough subjects to estimate the spread across test-subject groups, and report it. Compare networks paired on the same splits, never by means over different subjects. Average several training seeds. The reversal on S3 bounds the claim: a weak baseline is told apart from a strong network easily, and the subject effect matters when choosing between designs that are already competitive. The exploratory associations point at the sensing system: with many channels, hand-crafted features stay competitive, and with few recorded subjects the choice of network matters more.

The exploratory analyses agree with the decomposition and say where the subject effect sits. Within datasets, test groups spread accuracy more than networks do (Fig. 5). Validation on the training subjects overstates the person-disjoint score, on some datasets by tens of points, and ranks the networks differently (Fig. 6), so early stopping and model selection should hold out whole subjects. A test group is hard when its windows, class by class, lie far from the training subjects' in the space the classifier reads, a space that still identifies the training subjects (Fig. 9), whereas the temporal memories keep the scales they start with (Fig. 8). Together these suggest that, for comparable networks, the training subjects and how representations align them are more promising levers for accuracy on new people than another temporal operator.

The analysis covers one protocol, and three seeds per run. The S1 and S2 networks share one skeleton, so their components are controlled but not independent designs. S2's means were known before the registration. The 30-dataset results re-analyse published single runs and are correlational, and subjects and channels co-vary across datasets. The embedding analysis uses the seed-0 checkpoints of eleven datasets, and the validation analysis only the datasets whose runs recorded a validation score. The intervals resample test-subject groups and treat the networks as fixed: they say how precisely R is known for these networks, not for networks in general, and S3 shows that another population of networks can reverse it.

V. CONCLUSION

Under a fixed person-disjoint protocol, who is tested accounts for most of the variance of macro-F1 among comparable wearable HAR networks, as a pre-registered test on held-out datasets confirmed; across heterogeneous architectures the network dominates. Studies of wearable sensing systems should size and report their test subjects accordingly.

REFERENCES

[1] A. Bulling, U. Blanke, and B. Schiele, "A tutorial on human activity recognition using body-worn inertial sensors," *ACM Computing Surveys*, vol. 46, no. 3, pp. 33:1–33:33, 2014.

[2] N. Y. Hammerla and T. Plötz, "Let's (not) stick together: Pairwise similarity biases cross-validation in activity recognition," in *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. Osaka, Japan: ACM, 2015, pp. 1041–1051.

[3] J. W. Lockhart and G. M. Weiss, "Limitations with activity recognition methodology & data sets," in *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct Publication (UbiComp '14 Adjunct)*. Seattle, WA, USA: ACM, 2014, pp. 747–756.

[4] H. Qian, S. J. Pan, and C. Miao, "Latent independent excitation for generalizable sensor-based cross-person activity recognition," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 13, 2021.

[5] A. Stisen, H. Blunck, S. Bhattacharya, T. S. Prentow, M. B. Kjærgaard, A. Dey, T. Sonne, and M. M. Jensen, "Smart devices are different: Assessing and mitigating mobile sensing heterogeneities for activity recognition," in *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems (SenSys '15)*. Seoul, South Korea: ACM, 2015, pp. 127–140.

[6] M. Burzer, T. King, T. Riedel, M. Beigl, and T. Röddiger, "WHAR Arena: Benchmarking the state of the art in efficient wearable human activity recognition," arXiv:2606.13194, 2026.

[7] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Sepah, E. Raff, K. Madan, V. Voleti, S. E. Kahou, V. Michalski, D. Serdyuk, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 3, 2021.

[8] M. Burzer, T. King, T. Riedel, M. Beigl, and T. Röddiger, "WHAR datasets: An open source library for wearable human activity recognition," in *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*. Espoo, Finland: ACM, 2025, pp. 1315–1322.

[9] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *International Conference on Learning Representations (ICLR)*, 2022.

[10] A. Gu, A. Gupta, K. Goel, and C. Ré, "On the parameterization and initialization of diagonal state space models," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[11] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv:2312.00752, 2023.

[12] F. J. Ordóñez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition," *Sensors*, vol. 16, no. 1, p. 115, 2016.

[13] N. Y. Hammerla, S. Halloran, and T. Plötz, "Deep, convolutional, and recurrent models for human activity recognition using wearables," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2016.

[14] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[15] M. Christ, N. Braun, J. Neuffer, and A. W. Kempa-Liehr, "Time series FeatuRe extraction on basis of scalable hypothesis tests (tsfresh – A Python package)," *Neurocomputing*, vol. 307, pp. 72–77, 2018.

[16] J.-P. Eckmann, S. O. Kamphorst, and D. Ruelle, "Recurrence plots of dynamical systems," *Europhysics Letters*, vol. 4, no. 9, pp. 973–977, 1987.

[17] M. Brokate and J. Sprekels, *Hysteresis and Phase Transitions*, ser. Applied Mathematical Sciences. New York: Springer, 1996, vol. 121.

[18] H. Robbins and S. Monro, "A stochastic approximation method," *The Annals of Mathematical Statistics*, vol. 22, no. 3, pp. 400–407, 1951.

[19] H. D. Patterson and R. Thompson, "Recovery of inter-block information when block sizes are unequal," *Biometrika*, vol. 58, no. 3, pp. 545–554, 1971.

[20] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with Python," in *Proceedings of the 9th Python in Science Conference (SciPy 2010)*, Austin, TX, USA, 2010, pp. 92–96.

[21] C. A. Field and A. H. Welsh, "Bootstrapping clustered data," *Journal of the Royal Statistical Society Series B*, vol. 69, no. 3, pp. 369–390, 2007.

[22] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[23] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006. [Online]. Available: https://www.jmlr.org/papers/v7/demsar06a.html

[24] K. Zhang, B. Schölkopf, K. Muandet, and Z. Wang, "Domain adaptation under target and conditional shift," in *Proceedings of the 30th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 28, no. 3, 2013, pp. 819–827. [Online]. Available: https://proceedings.mlr.press/v28/zhang13d.html

[25] G. Alain and Y. Bengio, "Understanding intermediate layers using linear classifier probes," arXiv preprint arXiv:1610.01644, 2016.

[26] L. van der Maaten and G. Hinton, "Visualizing data using t-sne," *Journal of Machine Learning Research*, vol. 9, no. 86, pp. 2579–2605, 2008. [Online]. Available: https://www.jmlr.org/papers/v9/vandermaaten08a.html